

TECHNO WAREHOUSE PLANNING MODEL

COMPLETE SYSTEM ARCHITECTURE, BUSINESS LOGIC & ALLOCATION ENGINES

EXECUTIVE SUMMARY: The Techno Warehouse Planning & Allocation Model is an enterprise supply chain intelligence engine designed to optimize multi-warehouse inventory distribution, automate sales-driven ABC categorization, execute size & style-constrained inventory allocation from central mother hubs, and validate multi-stage projected stock availability across regional fulfillment centers.

System Core: Python Flask / Pandas Engine	Primary Mother Hub: Bagalur Hub
Regional Hubs: Bhiwandi, Gurugram, Kolkata	Allocation Engines: Dynamic Capping & Box-Wise Models
Data Ingestion: Sales & WMS CSV/Excel Files	Validation Pipeline: Raw, In-Transit, & Projected Stock

1. System Overview & Warehouse Architecture

The system coordinates inventory across **four strategic fulfillment locations** in India. Central inventory originates at the primary manufacturing/distribution center in Bagalur and is algorithmically pushed to regional fulfillment nodes:

Warehouse ID	Location Name	Role & Priority	Default Split Strategy
wms_bagalur	Bagalur Hub	Primary Mother Warehouse / Retention Center	Retains 60% base stock + excess uncapped stock
wms_bhiwandi	Bhiwandi (West)	Priority 1 Push Warehouse (P1)	Receives 20% base push allocation
wms_kolkata	Kolkata (East)	Priority 2 Push Warehouse (P2)	Receives 15% base push allocation
wms_ggn	Gurugram (North)	Priority 3 Push Warehouse (P3)	Receives 5% (or residual push) allocation

2. Sales-Driven ABC Inventory Classification Logic

To ensure high-performing SKUs receive maximum warehouse presence while long-tail inventory remains centralized, the system performs dynamic Pareto ABC analysis based on aggregate historical sales volume.

Mathematical Formula & Cumulative Sales Ranking:

- Aggregates sales by SKU: $Total\ Sales\ Qty = \sum (Sales\ Qty\ per\ SKU)$.
- Sorts SKUs descending by sales quantity.
- Calculates Cumulative Sales Percentage: $Cumulative\ \% = (\sum Cumulative\ Qty / Total\ System\ Sales) \times 100$.
- Assigns Category Code based on exact cutoffs:

Category	Cumulative Sales Cutoff	SKU Strategic Focus	Target Warehouse Coverage
Category A	0.00% – 50.00%	Top-selling velocity SKUs generating 50% of revenue.	Target 4 / 4 Warehouses (Full Regional Coverage)
Category B	50.01% – 75.00%	Medium-velocity core styles generating next 25% sales.	Target 2 to 3 Warehouses
Category C	75.01% – 100.00%	Slow-moving / long-tail SKUs generating bottom 25%.	Centralized in 1 Mother Warehouse (Bagalur)

3. Warehouse Stock Pivoting & Coverage Metrics

The analytics core dynamically pivots physical WMS inventory records across fulfillment locations. For each SKU, the system computes the **Warehouse Coverage Count (0 to 4 WHs)**:

Formula: *WH Count = Count of Warehouses where Total Available Quantity > 0*

Category Health Index: Calculates average warehouse presence per category: *Avg WH Coverage = $\sum(\text{WH Count per SKU}) / \text{Total SKUs in Category}$* .

Zero-Stock Gap Analysis: Identifies Category A SKUs with *Total Stock = 0* across all warehouses as critical out-of-stock risk alerts.

4. Allocation Engine 1: Dynamic Size & Style Capping

The Dynamic Allocation Planner processes available stock at Bagalur and determines optimal quantities to dispatch to Bhiwandi, Kolkata, and Gurugram using a **4-stage constraint hierarchy**:

Stage 1: Base Ratio Split: 60% retained at Bagalur. 40% Push Pool split as Bhiwandi=20%, Kolkata=15%, Gurugram=5%.

Stage 2: Size-Specific Maximum Capping: Pushed quantities per WH are capped against gender/size MAX limits. Excess is redirected to Bagalur.

Stage 3: Size-Specific Minimum Capping: If allocated qty < MIN cap (e.g. MIN=5), push is canceled (set to 0) and retained at Bagalur.

Stage 4: Aggregate Style-Level NS & PS Capping: Sizes grouped into Normal (NS: S–2XL) and Plus (PS: 3XL–5XL). If aggregate style-color push to a WH is < overall MIN cap (e.g. Mens NS Min = 25 pcs), ALL size allocations for that style to that WH revert to Bagalur.

Size-Level Capping Reference Table (Mens vs Womens):

Category / Size	Kolkata Caps (MIN / MAX)	Bhiwandi Caps (MIN / MAX)	Gurugram Caps (MIN / MAX)
Mens S / M	Min: 5 Max: 15–20	Min: 5 Max: 15–20	Min: 5 Max: 10–15
Mens L / XL	Min: 5 Max: 30	Min: 5 Max: 30	Min: 5 Max: 20
Mens 2XL	Min: 5 Max: 20	Min: 5 Max: 20	Min: 5 Max: 10
Mens 3XL (PS)	Min: 3 Max: 10	Min: 3 Max: 10	Min: 0 Max: 0 (No Push)
Mens 4XL / 5XL (PS)	Min: 2 Max: 5	Min: 2 Max: 5	Min: 0 Max: 0 (No Push)
Womens S to 2XL	Min: 3 Max: 10–20	Min: 3 Max: 10–20	Min: 3 Max: 5–15
Womens 3XL to 5XL	Min: 0 Max: 0 (No Push)	Min: 0 Max: 0 (No Push)	Min: 0 Max: 0 (No Push)

Overall Style-Level NS & PS Minimum Thresholds:

Group Type	Size Range Included	Kolkata Min Threshold	Bhiwandi Min Threshold	Gurugram Min Threshold
Mens NS Group	S, M, L, XL, 2XL	25 Pieces	25 Pieces	25 Pieces
Mens PS Group	3XL, 4XL, 5XL	7 Pieces	7 Pieces	0 Pieces
Womens NS Group	S, M, L, XL, 2XL	3 Pieces	3 Pieces	3 Pieces

5. Allocation Engine 2: Box-Wise Allocation Model

For standardized carton-based fulfillment, the portal provides a specialized **Box-Wise Allocation Engine**. It converts unit allocations into exact master box counts (Standard = 10 pcs/box):

- Total Master Boxes:** *Total Boxes = Grand Total Qty / Per Box Qty (rounded to 2 decimals)*
- Kolkata Alloc Qty:** *Kolkata Alloc = MIN(Size Rule Cap, Grand Total × 10%)*

3. **Bhiwandi Alloc Qty:** *Bhiwandi Alloc = MIN(Size Rule Cap, Grand Total × 10%)*
4. **Gurugram Alloc Qty:** *Gurugram Alloc = MIN(Size Rule Cap, Grand Total × 6%)*
5. **Bagalur Retained Stock:** *Bagalur Retained = Grand Total - (Kolkata + Bhiwandi + Gurugram Pushed Qty)*

Size Code	Rule Mapping	Kolkata Max Rule	Bhiwandi Max Rule	Gurugram Max Rule
S	SML Rule	15 pcs	15 pcs	10 pcs
M	MED Rule	20 pcs	20 pcs	15 pcs
L / XL	LAR / XLR Rule	30 pcs	30 pcs	25 pcs
2XL	2XL Rule	20 pcs	20 pcs	15 pcs
3XL / 4XL / 5XL	Plus Size Rules	5 pcs	5 pcs	2 pcs
06Y – 10Y (Kids)	Junior Size Rules	8 pcs	6 pcs	6 pcs
12Y – 16Y (Kids)	Senior Kids Rules	10 pcs	8 pcs	6 pcs

6. Warehouse Validation & 3-Stage Pipeline Projection

The Warehouse Validation tab reconciles physical warehouse stock against planned allocation transfers and incoming factory shipments across **three comparative pipeline stages**:

Pipeline View Stage	Stock Included in Calculation	SKU Coverage Impact Target
Raw Stock Only	Physical stock in warehouse WMS	Establishes baseline regional stock gaps.
Raw + In-Transit	Physical Stock + Incoming Supply Shipments	Highlights incoming relief before allocation.
Fully Projected	Allocated Pushed Stock + In-Transit Shipments	Validates shift of Category A SKUs towards 4 / 4 WH Coverage .

7. REST API Endpoints & Technical Data Matrix

Endpoint Route	Method	Function & Payload Output
/api/summary	GET	Returns system total SKUs, total sales, ABC breakdown, stock per warehouse, and overall WH distribution.
/api/details	GET	Returns paginated SKU records with sales rank, sales qty, category, size/color metadata, and per-WH quantities. Supports search & WH filter.
/api/run_allocation	GET	Executes Dynamic Capping Planner across inventory dataset; outputs summary statistics and exports allocation_output.csv.
/api/run_boxwise_allocation	GET	Executes Box-Wise Allocation Engine; calculates allocated units, box counts per WH, and exports allocation_output.csv.
/api/validation	GET	Returns 3-stage validation records comparing Current, In-Transit, and Projected stock across all 4 warehouses.
/api/download_allocation	GET	Downloads the generated allocation output CSV file for execution by warehouse operations.

DOCUMENT CONTROL & SYSTEM AUTHORIZATION: This document specifies the complete algorithmic logic for the Techno Warehouse Planning Model. Approved for operational deployment and system audit.